

Do teams need both hands? An analysis of team process ambidexterity and the enabling role of information technology

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ABSTRACT

As a core organizational resource, business processes are vital for organizational teams. To deal with today's volatile business environment, organizations need to be ambidextrous in terms of process capabilities. However, little is known about how process ambidexterity, process standardization, and process agility, are enabled by information technology (IT) and related to team-level activities. To fill this gap in the literature, we conducted a field study based on 160 teams of 1081 individuals from seven companies in South Korea. Our results show that IT enables both process standardization and agility, and that a team's process ambidexterity has a positive effect on inter-team coordination and team innovation, which in turn have a direct impact on team performance. Our findings highlight the importance of process ambidexterity by investigating the enabling role of IT and its outcomes in a team. Our results offer theoretical and practical implications from the perspective of team process ambidexterity.

1. Introduction

Business processes influence every behavior of organizational actors, including teams. By designing and executing processes, organizations accomplish their desired purposes and objectives (Ray, Barney, & Muhanna, 2004). As business processes have become valuable assets leading to improvements in organizational performance, organizations have introduced a variety of methods for business process management (BPM), such as process standardization, which is a proven long-term process management method. Standardized processes enable companies to align organizational resources with organizational units to improve overall operational efficiency and accuracy (Lee, DeLone, & Espinosa, 2010). Furthermore, companies can benefit from process standardization by using well-defined templates to guarantee transparent and uniform process execution (Wüllenweber, Beimborn, Weitzel, & König, 2008); minimize variances and uncertainties (Tregear, 2015); and optimize time, cost, and quality (Münstermann, Eckhardt, & Weitzel, 2010).

Although well-established standards can certainly improve the operational efficiency of business processes, adapting to the ever-changing business environment has become a critical capability for organizations to survive; thus, business processes need to be adaptable to changes (Chen et al., 2014; Shukla, 2020). This capability is called process

agility and refers to the ability to adjust business processes in response to changes (Tallon, 2008). The importance of process agility has been growing, as proactive response to market changes has strong relevance to business performance (Raschke, 2010) and how firms deal with the market environment is critical to their survival (Van Oosterhout, Waarts, & van Hillebergersberg, 2006).

However, having only one good quality is not sufficient for creating organizational values for survival. Managers and researchers have pointed out that pursuing two different, or ambidextrous, qualities is important for business performance despite being a challenging task (He & Wong, 2004). With respect to process management, managing the ambidexterity of process capabilities can be a critical task, as business processes should be able to both enhance the efficiency of execution and adapt to changes in the external environment (Lee et al., 2010). Even though having different process capabilities—for instance, process standardization and process agility—simultaneously is difficult, both are essential for team operations. Although the importance of process ambidexterity has grown (Rosemann, 2014), the issue has not received sufficient empirical attention.

Information technology (IT) is worth noting for enabling business processes, as it plays an important role in organizational processes with respect to transformation of resources into capabilities, which is critical to creating an organization's competitive advantages (Kohli & Grover,

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2008). As an important enabler of business processes by leveraging organizational resources (Melville, Kraemer, & Gurbaxani, 2004), IT can enhance process capabilities by offering diverse technological support and functionalities (Chen et al., 2014). Moreover, it offers capabilities that can help teams achieve competitive advantage, which ultimately leads to performance improvements (Choi, Lee, & Yoo, 2010), with such improvement in performance being achieved through intermediary processes (Melville et al., 2004). In other words, business processes play a linking role between IT and the performance of an entity.

The impacts of process ambidexterity are not limited only to the organization level, as ambidexterity also influences teams. As a typical characteristic of contemporary organizations, a team-based structure improves efficiency in executing organizational tasks (Shin, Kim, Lee, & Bian, 2012); thus, organizations need to design business processes that maximize the benefits of team structure. Teams often need different process capabilities simultaneously, which necessitates process ambidexterity. On the one hand, innovation in teams requires process agility, as innovation is highly correlated with exploiting opportunities (Choi & Chang, 2009). On the other hand, inter-team coordination can be related to process standardization because communications between teams follow pre-determined protocols (Davison, Hollenbeck, Barnes, Slesman, & Ilgen, 2012). Nevertheless, we have a limited understanding of process ambidexterity in teams. To fill this gap in the literature, we investigate process ambidexterity within the context of teams and analyze how it is linked with its antecedents and consequences. In other words, this study attempts to answer the following research questions. *In a team context, (1) how does IT support affect process ambidexterity? and (2) how does process ambidexterity affect different team-level behaviors such as inter-team coordination and team innovation?*

To answer the above research questions, we collected a sample of 160 teams from seven companies in South Korea. The analysis results show that IT support positively affects process ambidexterity, and that both process standardization and agility have positive impacts on inter-team coordination and team innovation, which lead to high team performance. Based on our empirical findings, this study contributes to the literature in several ways. By analyzing process ambidexterity, which consists of process standardization and process agility, we contribute to the BPM literature by demonstrating the importance of the dual capacity of process capability and its empirical validation in a team context. Additionally, this study contributes to the IT and process literature by identifying the role of IT in enabling process ambidexterity. Moreover, we reveal that process ambidexterity can influence different team-level behaviors simultaneously, thus hinting at the nomological net through process ambidexterity in a team context.

The remainder of this paper is structured as follows. First, we elaborate on the concept of process ambidexterity through a literature review. Then, we present the hypotheses, research design, and findings. Finally, we offer the theoretical and practical implications of our findings and directions for future research.

2. Literature review and hypotheses

2.1. Process ambidexterity

Organizational units are designed for a specific function. However, it is inevitable to confront conflicts when chasing multiple goals. Therefore, researchers have adopted the concept of ambidexterity to investigate such dynamics (Chi, Zhao, George, Li, & Zhai, 2017; de Guinea & Raymond, 2020). Ambidexterity refers to an organizational ability to simultaneously pursue and achieve two dissimilar goals. A well-known example of ambidexterity is exploration and exploitation (March, 1991), where exploration is associated with the search for new competencies, while exploitation is related to the use of existing competencies. In addition, ambidexterity has been studied in various forms

including evolution and revolution, rigor and flexibility (Lee, DeLone, & Espinosa, 2006), and alignment and adaptability (Raisch & Birkinshaw, 2008).

The BPM literature has consistently pointed out the clash between efficiency and flexibility with regard to business processes. On the one hand, business processes need to follow unified patterns and rules for enhanced efficiency. Organizations try to standardize processes, as consistency and repeatability of operations create values by aligning organizational resources with operations (Münstermann et al., 2010). Additionally, standardized processes save overhead and transaction costs, assure a certain level of procedural quality with fewer errors (Wüllenweber et al., 2008), and exploit economies of scale, thus leading to better performance (Ramakumar & Cooper, 2004). On the other hand, being flexible and responsive to external challenges becomes important in the ever-changing business circumstances. Processes need to be flexible enough to deal with unpredictable changes (Van Oosterhout et al., 2006). By utilizing agile processes, organizations can reconfigure resources, employ emerging technologies, and seize opportunities for innovation and strategic movements (Chen et al., 2014; Mathiassen & Pries-Heje, 2006; Raschke, 2010).

Process ambidexterity is important, as business processes need to be not only aligned, but also adaptable (Lee, Espinosa, & DeLone, 2013). However, enhancing a process capability could harm another. For instance, imposing strict rules on standardization can squash the flexibility and agility of a team's process (Trkman, 2010). On the contrary, a team cannot standardize their operations when it is engrossed in finding new opportunities due to the highly uncertain and flexible nature of such activities (Lyytinen & Rose, 2006). Therefore, standardization and agility can be factors that constitute process ambidexterity (Reif, Kugler, & Brodbeck, 2018) as both are important for process capability; however, it is hard to achieve the two simultaneously. Researchers have investigated the effects of process ambidexterity on firm performance (Xie, Ling, & Zhang, 2011), coordination effectiveness (Lee et al., 2013), and system performance in distributed development (Lee et al., 2010). However, empirical findings cannot provide a generalizable understanding, and are mainly focused on its consequences in an organizational context.

2.2. IT and process capability

The literature acknowledges the value of IT in enhancing diverse capabilities and supporting successful execution of business processes (Chen et al., 2014; Mithas, Ramasubbu, & Sambamurthy, 2011). IT can help locate and transfer necessary organizational resources, resolve dependencies between tasks, and increase the efficiency of resource use (Bala, Massey, & Montoya, 2017). These enhancements are not meant to support a single process capability. From the viewpoint of process ambidexterity, there can be a multitude of relationships between IT and process capability. However, there is a lack of research that addresses which process capabilities can be improved by IT, especially in a team context. Teams often need different process capabilities simultaneously, as they execute a variety of business processes and activities, which necessitates process ambidexterity. Therefore, understanding the relationship between IT and process ambidexterity in teams can achieve better team process management and leverage the team-based structure. To fill the research gap, we hypothesize relationships between IT and two capabilities that constitute process ambidexterity, namely, standardization and agility.

IT plays an important role in process standardization for integrating, reorganizing, and revising processes (Ross & Vitale, 2000), and processes become easily measurable when proper IT support is provided (Rosemann & vom Brocke, 2015). The enhanced trackability of business processes can lead to a better understanding of how the team's tasks are inter-related and executed in terms of the participants, the person in charge, the criteria for success, and the technological requirements (Mao, Liu, Zhang, & Deng, 2016; vom Brocke, Zelt, & Schmiedel, 2016).

Furthermore, variances in process outcomes can be minimized when they can be measured and evaluated, and accordingly, uniform executions of processes become possible. As understanding processes and related resources is a prerequisite for process standardization, an enhanced knowledge of business processes can help standardize processes. Therefore, a higher degree of IT support is more likely to lead to process standardization (Beimborn, Gleisner, Joachim, & Hackethal, 2009). Hence, we postulate the following hypothesis:

H1. IT support has a positive effect on process standardization in teams.

As the uncertainty in the business environment increases, a team's operation is required to advance beyond pursuing efficiency only. The literature suggests that IT can provide both capabilities for standardization and agility by offering flexible ways of operation (Queiroz, Tallon, Sharma, & Coltman, 2018; Tallon & Pinsonneault, 2011) and enhancing the reactivity in response to environmental changes (Shukla, 2020; vom Brocke et al., 2016). In the light of information management, proper IT support can boost the speed of information storing, processing, analyzing, and decision-making (Sidorova, Torres, & Al Beayez, 2015). Furthermore, when a team has an easy access via IT systems to organizational resources regardless of time and location, all possible resources can be utilized to respond to changes, significantly enhancing its process to respond to external environmental changes, and shortening the time to make critical strategic decisions (Huang, Pan, & Ouyang, 2014; Park, El Sawy, & Fiss, 2017). Therefore, a higher degree of IT support is more likely to lead to process agility. Hence, we propose:

H2. IT support has a positive effect on process agility in teams.

2.3. Process ambidexterity and team-level activities

Inter-team coordination refers to the efforts by team members to adjust their actions to those of other teams to manage the interdependencies among them (Hoegl, Weinkauff, & Gemuenden, 2004), with proactive coordination among teams being critical to achieving organization-level outcomes (Cha, Kim, Lee, & Bachrach, 2015). Collaboration between teams can create synergies and shared knowledge that can be applicable to other contexts (Olabisi & Lewis, 2018).

As a sequence of processes often requires the involvement of several teams, coordination between teams is critical. To coordinate efficiently, teams need predefined and shared operational methods, (Davison et al., 2012), including common language, understanding of schedules and procedures, and conflict management protocols to address problematic issues (Lee et al., 2013; Wüllenweber et al., 2008). As executing processes in a standardized way can deal with the abovementioned factors, it plays an important role in coordinating teams (Reif et al., 2018). Hence, we propose:

H3. Process standardization has a positive effect on inter-team coordination.

Inter-team coordination aligns team members' activities with those of other teams to deal with organization-wide dependencies (de Vries, Walter, Van der Vegt, & Essens, 2014), with teams reconfiguring their resources dynamically to meet other teams' needs and receiving feedback from them. Process agility benefits teams in terms of handling communication contingencies and generating multiple alternatives (Athey & Schmutzler, 1995), and enables them to respond to unexpected challenges and coordinate with other teams (Chen et al., 2014), thus being helpful in coordination among teams. Additionally, the boundary spanning roles in agile teams (Marrone, Tesluk, & Carson, 2007) further encourage inter-team coordination. Hence, we propose:

H4. Process agility has a positive effect on inter-team coordination.

Innovation ensures the organization's long-term survival (Anderson, Potočník, & Zhou, 2014). As innovation requires group-level learning and knowledge transfer under restrictive circumstances, it inherently

occurs in teams (Edmondson, 2002), with group-level processes having the ability to transform the members' creativity into innovative outcomes (Taggar, 2002). However, standardization is an impediment to innovation because innovative activities seem risky and abnormal compared to the existing norms (Manders, de Vries, & Blind, 2016). Additionally, the rule-complying team climate hinders members from thinking outside the box and suggesting alternatives (Choi, Narasimhan, & Kim, 2016). Moreover, routinized processes lead individuals to think that the current way of work is the best they can perform (Benner & Tushman, 2002), as highly standardized processes can reduce flexibility and discourage innovative activities (Choi, Anderson, & Veillette, 2009). Therefore, the innovative team climate, essential for encouraging innovative actions, declines due to process standardization (Choi, 2007). Hence, we propose:

H5. Process standardization has a negative effect on team innovation.

Agility is vital to competitive performance in the contemporary business environment. Organizations can rapidly and flexibly respond to business challenges and access required resources when their agile processes are ready to be utilized (Sambamurthy, Bharadwaj, & Grover, 2003). Thus, teams in agile organizations foster innovativeness so that they can be responsive to volatile market dynamics (Mathiassen & Pries-Heje, 2006), as they capitalize on available resources in a given circumstance to seize opportunities. Therefore, process agility enables teams to perform innovatively. Hence, we propose:

H6. Process agility has a positive effect on team innovation.

2.4. Team performance

Although the cost of coordination with other teams is not negligible, the advantages of inter-team coordination are worthwhile. From an organization's perspective, its teams should cooperate with each other to meet its holistic goals and vision. Inter-team coordination has been recognized as an important factor affecting team performance, with prior studies showing that inter-team coordination between team members is positively associated with both team performance and overall organizational performance (Hoegl et al., 2004; Lee, Min, & Lee, 2017). A team can gain additional resources when its members actively participate in boundary-spanning activities (i.e., inter-team coordination), resulting in better performance (de Vries et al., 2014; Marrone et al., 2007). Inter-team coordination also provides the opportunity to obtain new perspectives and alternative ideas from external experts (Sethi, 2000). Additionally, inter-team coordination may be accompanied by assistance from other teams and their members for resolving complexities and uncertainties (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). In contrast, poor inter-team coordination can result in unnecessary delays and significant reworking (Joglekar, Yassine, Eppinger, & Whitney, 2001). Therefore, we hypothesize:

H7. Inter-team coordination has a positive effect on team performance.

Innovation often leads to better performance by enhancing the competitive advantages of an organization (Ali, Kan, & Sarstedt, 2016; Huang, Wu, Lu, & Lin, 2016), with IT-enabled innovation becoming a key driver of organizational performance by enabling the organization to seize new opportunities (Talke, Salomo, & Kock, 2011; Yunis, Tarhini, & Kassar, 2018). We suggest that this linkage is also the case at the team level, as the positive relationship between innovation and performance is empirically shown in research and development teams (Bain, Mann, & Pirola-Merlo, 2001). Innovation in teams is likely to lead to better team performance, as team members are encouraged to apply new ideas and procedures proactively. Thus, innovative teams increase the capabilities of an organization to solve challenging business problems. Hence, we propose:

H8. Team innovation has a positive effect on team performance.

Table 1
Summary of the sample profile.

Individual (N = 1,081)			
Characteristics		Frequency	Percent
Job Position	Employee	495	45.8%
	Chief	208	19.2%
	Manager	112	10.4%
	Director	185	17.2%
	Managing Director	81	7.4%
Age (years)	20–30	388	35.9%
	31–40	382	35.4%
	41–50	261	24.1%
	51+	50	4.6%
Company Tenure (years)	0–2	170	15.7%
	2–5	265	24.5%
	5–10	271	25.1%
	10–15	94	8.8%
	15–20	146	13.5%
	20+	135	12.4%
Gender	Male	739	68.4%
	Female	342	31.6%
Team (N = 160)			
Function	Manufacturing	105	65.6%
	Non-manufacturing	55	34.4%
Team Size (people)	5–10	61	38.1%
	10–14	26	16.2%
	15–20	28	17.5%
	20–25	15	9.4%
	25+	30	18.8%

3. Method

3.1. Data and sample

We collected data from seven Korean firms in diverse industries such as health care, chemical engineering, construction, system integration, heavy industry, financial industry, and petrochemical industry. A web-based survey was sent via e-mail to each company. Afterwards, questionnaires were distributed by cooperators inside the companies. All the respondents voluntarily participated in the online survey. When the questionnaires were completed, the insider co-operators collected the data and sent it to us. The total number of respondents was 2471, belonging to 196 teams. For team-level analysis, the inter-rater agreement (r_{wg}) needs to be evaluated to validate the within-group consensus (James, Demaree, & Wolf, 1984). First, we eliminated teams that had fewer than three members because they do not adequately represent their teams' characteristics. Subsequently, we calculated the interclass correlation coefficient (ICC) (Shrout & Fleiss, 1979). The final data set for further analysis included 160 teams with 1081 individuals. Our sample characteristics are presented in Table 1.

3.2. Measures

All the items designed to capture the team-level characteristics were measured using a 7-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree"). The detailed items and questions for the constructs are listed in Appendix A. Five items were designed to measure the team's overall performance in relation to its achievements, which were adopted from Hoegl et al. (2004). We developed four items for team innovation based on Lovelace, Shapiro, and Weingart (2001). The five items used to quantify the inter-team coordination were also adopted from Hoegl et al. (2004). These items were designed to reflect the operational characteristics of the teams as well as the quality of the inter-team coordination.

Measuring process agility involves quantifying the degree of a team's ability to respond to external changes and make proper transitions. Four items to measure process agility were adopted from Lee et al. (2013). The items used for process standardization measured the

extent and consistency of the standards with respect to common tasks and communication. Four items were used to assess process standardization (Lee et al., 2013). We measured IT support using four items developed by Lee and Choi (2003). These items measure the perceived IT support for the respondents' teams in terms of communication, collaboration, search, access, and storage of information. Of the five items developed by Lee and Choi (2003), we excluded one item related to simulation because of the mismatch of context.

We controlled for variables that can affect team performance, such as team type, team size, and company (Choi et al., 2010; Lee, Min, Kwak, Pee, & Lee, 2019). The team type is divided into manufacturing and non-manufacturing, while the team size refers to the number of employees working in a specific team.

4. Results

To use the individuals' responses for further analysis, the responses were aggregated by extracting the mean scores. To test our hypotheses, we used the partial least squares (PLS) method because of its minimal restrictions on the distribution of residuals and advantage in testing a small sample size (Chin, 1998). As the size of the final sample is relatively small ($n = 160$), PLS is a suitable choice (Lee et al., 2019). The results from the PLS structural equation model were analyzed in two parts. First, we examined the measurement model to verify certain issues related to the constructs, including convergent validity, discriminant validity, and reliability. Subsequently, we examined the structural model to test the proposed hypotheses and paths.

4.1. Measurement model

To assess the construct validity of the measures, we conducted a confirmatory factor analysis (CFA), with its results presented in Table 2. Even though cross-loading can be observed, all the items have greater loadings on their constructs compared to the loadings on other constructs. The Cronbach's alpha and composite reliability (CR) values for all the constructs are reported in Table 2, with all of them meeting satisfactory levels (Hair, Black, Babin, Anderson, & Tatham, 2006). Table 3 presents the correlations between the constructs whose diagonal values are the square roots of the average variance extracted (AVE). The diagonal values are greater than the non-diagonal construct correlations of the same rows and columns (Fornell & Larcker, 1981). To investigate possible high correlations between variables, we calculated the variance inflation factor (VIF), finding that the range of VIF is from 1.28 to 3.00, showing an acceptable value (Hair et al., 2006). Combined with the CFA results, we conclude that the constructs used in this study demonstrate both convergent and discriminant validity. During this process, we omitted three items for a better reliability, one related to team performance and two related to inter-team coordination.

In addition to the aggregation procedures, we examined the possibility of common method bias (CMB) using two approaches. First, we conducted the Harman's single factor test (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). According to the test, our model is not affected by CMB, as the first factor of an exploratory factor analysis accounts for less than 50 % of the variance (Podsakoff & Organ, 1986). We also investigated CMB using the method by Liang, Saraf, Hu, and Xue (2007) (see Appendix B). Compared to the average variance explained by the method (0.01), the average variance of the indicators explained by the constructs is 0.91, with the ratio between them being 91:1. Therefore, we conclude that the validity of this study is not threatened by common method bias.

4.2. Structural model

Our PLS results are shown in Fig. 1. The effects of IT support on process standardization and process agility are significant, supporting

Table 2
Confirmatory factor analysis and reliability measures.

Construct	Item	Factor 1	2	3	4	5	6	Cronbach's α	CR
IT Support	ITS1	0.91	0.09	0.07	0.14	0.10	0.17	0.97	0.98
	ITS2	0.92	0.10	0.11	0.20	0.14	0.11		
	ITS3	0.90	0.13	0.21	0.17	0.06	0.14		
	ITS4	0.93	0.13	0.16	0.16	0.04	0.13		
Process Standardization	PS1	0.18	0.78	0.24	0.23	0.18	0.30	0.98	0.98
	PS2	0.13	0.89	0.19	0.18	0.17	0.18		
	PS3	0.13	0.88	0.19	0.20	0.17	0.22		
	PS4	0.11	0.87	0.19	0.20	0.20	0.20		
Process Agility	PA1	0.05	0.87	0.20	0.22	0.13	0.22	0.98	0.98
	PA2	0.15	0.27	0.84	0.22	0.22	0.17		
	PA3	0.19	0.21	0.86	0.22	0.22	0.18		
	PA4	0.16	0.25	0.84	0.19	0.21	0.25		
Team Innovation	TI1	0.17	0.22	0.85	0.24	0.17	0.25	0.97	0.98
	TI2	0.20	0.28	0.21	0.76	0.24	0.27		
	TI3	0.23	0.25	0.24	0.82	0.13	0.22		
	TI4	0.29	0.27	0.27	0.79	0.17	0.24		
Inter-team Coordination	TC2	0.24	0.30	0.27	0.79	0.17	0.25	0.93	0.96
	TC3	0.03	0.13	0.21	0.25	0.85	0.09		
	TC4	0.15	0.25	0.18	0.11	0.86	0.20		
Team Performance	TP2	0.16	0.27	0.26	0.13	0.82	0.20	0.96	0.97
	TP3	0.19	0.27	0.27	0.33	0.19	0.76		
	TP4	0.23	0.30	0.24	0.29	0.13	0.77		
	TP5	0.13	0.31	0.16	0.17	0.14	0.84		
		0.15	0.22	0.22	0.17	0.17	0.85		

Table 3
The correlation matrix.

Construct	Mean	S.D.	ITS	PS	PA	TI	TC	TP
ITS	4.45	0.86	0.96					
PS	5.19	0.66	0.34	0.95				
PA	5.22	0.67	0.41	0.56	0.97			
TI	4.94	0.79	0.51	0.61	0.62	0.94		
TC	4.90	0.71	0.31	0.51	0.55	0.51	0.95	
TP	5.36	0.73	0.43	0.62	0.59	0.65	0.48	0.94

H1 and H2. Process standardization and process agility are positively related to inter-team coordination and team innovation except for the link between standardization and innovation, supporting H3, H4, and H6. We find an unexpected positive and significant linkage from process standardization to team innovation; thus, H5 is not supported. As predicted, both inter-team coordination and team innovation are found to have positive impacts on team performance, supporting H7 and H8.

With regard to the control variables, two variables show a significant relationship with team performance, with manufacturing teams showing a higher level of team performance. In addition, a system integration company shows a lower level of team performance compared to other companies. The effect of team size remains insignificant.

5. Discussion

This study focuses on process standardization and agility as constituents of process ambidexterity and explores their roles in the behaviors of teams by answering the following questions. How does IT support affect process ambidexterity? and how does process ambidexterity affect different team-level behaviors such as inter-team coordination and team innovation? Based on the sample of 160 teams, our analysis indicates several points. First, we find that IT is an important enabler of team process ambidexterity, namely, IT enhances both process standardization and process agility. While process standardization and process agility may have dissimilar goals and play a role as different hands, teams can standardize work processes and improve agility to adapt to changes simultaneously with support of IT. This result enriches previous findings that IT can strengthen various processes' capabilities (Rosemann & vom Brocke, 2015) to be ambidextrous and be able to handle disparate processes in a team context.

Furthermore, we find a positive linkage from process ambidexterity to team-level behaviors, with process agility having a positive influence on both team-level behaviors. As we hypothesized, the ability of agile processes to adapt to changes is crucial for teams to coordinate and innovate their activities and ideas, thus enhancing both team innovation and coordination with other teams (Chen et al., 2014; Mathiassen

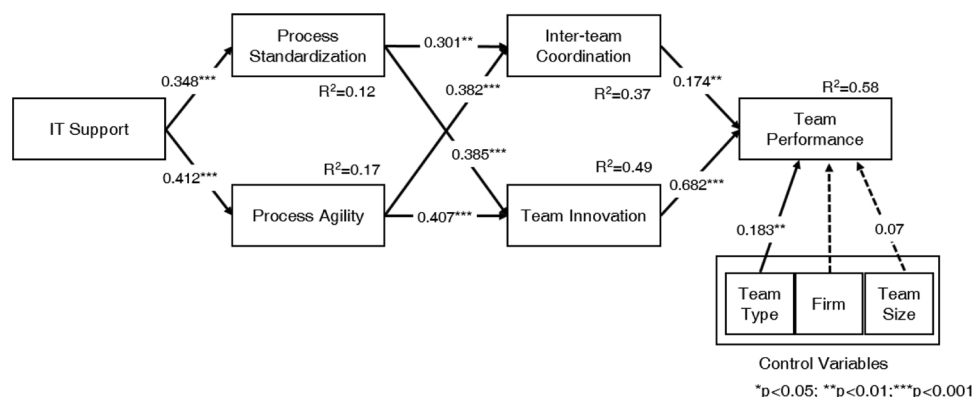


Fig. 1. The analysis results.

& Pries-Heje, 2006). In case of process standardization, it also has a positive influence on inter-team coordination and team innovation (Reif et al., 2018). However, the positive relationship between process standardization and team innovation contradicts our expectation. A plausible explanation is that the standardized ways of operation contribute to innovation through compatibility (Fixson & Park, 2008), which promotes openness and collaboration between internal employees and external partners, as team members become less likely to resist changes when they have a common understanding of product components and platforms (Funk & Luo, 2015). Standardization is helpful in innovations, as process standardization can enable team members to better understand existing technologies, notice potential problems, and pursue innovative ideas (Wright, Sturdy, & Wylie, 2012). Interestingly, the effects of agility on team-level behaviors are found to be greater than those of standardization. It seems that the importance of agility has been increasing due to its significance in today's rapidly evolving business environment.

As expected, both inter-team coordination and team innovation have positive effects on team performance, with the effect of team innovation being greater than that of inter-team coordination. The increasing importance of innovation may also be due to the volatile business landscape. Additionally, by revealing the hypothesized relationships, we believe that this study might provide a hint at the nomological net between IT support and team performance through ambidextrous processes and teams' activities. Our results confirm that the previous findings in an organizational context, suggesting that a consistent integration of IT and processes is necessary to create business value (Peng, Quan, Zhang, & Dubinsky, 2016), are also reproduced in a team context.

5.1. Theoretical implications

This study has several implications for research. First, our field study investigates the dual capacity of team process ambidexterity, whereas the majority of prior studies on process capabilities has examined each process capability separately, standardization (Münstermann et al., 2010) or agility (Chen et al., 2014), rather than simultaneously in an organizational context. To close this gap in the literature, our study explores the complementary role of process ambidexterity with its linkages from IT support and team-level behaviors, to team performance. To the best of our knowledge, this study is the first to provide deeper insights into the understanding of process ambidexterity by empirically identifying its roles in a team context.

Second, this study clearly demonstrates the role of IT in fostering team process ambidexterity. While the enabling role of IT is quite traditional (Melville et al., 2004), this study confirms an enabling role of IT in both standardization and agility from the perspective of process ambidexterity. In other words, IT can enable ambidextrous and seemingly disparate process capabilities, rather than just enhance a single capability. This finding is an extension of the previous finding that processes should be aligned with IT within the organizational context (Coltman, Tallon, Sharma, & Queiroz, 2015). Our study contributes to the IT literature by reexamining the role of IT in business processes and management, and identifying its importance in teams. Moreover, we also contribute to the BPM literature by identifying IT as an important enabler of team process ambidexterity, while previous studies on BPM have mainly focused on the consequences of processes.

Third, this study reveals the positive linkages between process ambidexterity and team-level behaviors, including innovation and coordination. By considering standardization and agility simultaneously in teams, our study analyzes the complementary impact of process ambidexterity on team activities. Moreover, this study hints at the nomological net from IT support to team performance focusing on ambidextrous processes. Based on our findings, scholars can deepen our understanding of process ambidexterity and its roles in the context of team and its operation.

5.2. Implications for practice

This study offers several practical implications. First, our analysis suggests that managers should focus on facilitating process ambidexterity in teams. Managers may think that a team behavior depends on a particular process capability. For instance, team innovativeness seems to have a close relationship with process agility, while inter-team coordination seems to rely on process standardization. However, our results show that both standardization and agility are critical to enhancing coordination and innovation. Therefore, to improve various team behaviors, it is important for managers to make a team's process capability ambidextrous. Through process ambidexterity, managers can make their teams more versatile, leading to a higher team performance. Thus, although simultaneously enabling both disparate and ambidextrous processes is challenging, managers have to pursue process ambidexterity.

Furthermore, managers should focus on their IT support to ensure a high team process ambidexterity. While the enabling role of IT is well-known, this study goes one step further by presenting that IT can enhance different process capabilities simultaneously. IT support is found to play a significant role in reinforcing process standardization and process agility, forming process ambidexterity. IT support can help process standardization in terms of the enhanced utilization of organizational resources and process management tools. Meanwhile, IT support can improve process agility by offering a greater flexibility to a team. A lack of IT support results in an inferior team process ambidexterity, which potentially degrades the general execution of the team processes, and leading to a lower team performance. Managers should carefully formulate strategies to utilize IT for their team members to innovate and coordinate effectively, and strengthen the IT infrastructure to support team processes.

Moreover, our findings provide a guideline for a better understanding of process capability allocation. When managers have a limited capacity to pursue both processes, they should realize the increasing importance of process agility, as its effect on coordination and innovation is greater than that of process standardization in isolation. Moreover, with regard to teams' activities, while managers have to pursue both coordination and innovation, they need to focus more on innovation rather than coordination, as innovation can lead to a higher performance. Managers should be aware that process agility and team innovation are more effective in responding to today's ever-changing business situation.

5.3. Limitations and future research directions

This study has some areas for further research. First, we employed a cross-sectional analysis. This research design covers limited aspects of process ambidexterity and its effects on teams, while a research with longitudinal data will provide a deeper understanding of ambidextrous processes. For example, while process standardization can enhance or hamper process agility and vice versa, (Gattiker, Chen, & Goodhue, 2005; Trinh, Molla, & Peszynski, 2013), our sample cannot verify these relationships; thus, we excluded them from our research model. However, longitudinal data can consider these process dynamics and provide a broader understanding to academia and practice.

Second, the process management literature has shown that various contextual factors can affect BPM (vom Brocke et al., 2016). For example, as the task characteristics are important factors in team operation, a better understanding of process ambidexterity may be possible by the incorporation of task characteristics into the current model. In a similar vein, studies with the contextual factors of team can provide a further integrated understanding of process ambidexterity.

Lastly, although this study considers IT support as a single dimension, investigating IT subdimensions can enhance our understanding of the relationship between IT and process ambidexterity. Researchers have suggested that different components of IT can have different

effects on diverse components of business contexts (Chi et al., 2017; de Guinea & Raymond, 2020; Mao et al., 2016). Thus, revealing the interactions between the subdivisions of IT and process ambidexterity can significantly expand this study.

6. Conclusions

As business processes are essential resources for organization to deal with volatile business environments, teams necessitate ambidexterity in process capabilities. However, there is a lack of empirical understanding of how team process ambidexterity, process standardization, and process agility, are enabled by IT and related to team-level activ-

ities. To fill this gap in the literature, this study establishes a research model from IT support to team performance by focusing on team process ambidexterity, and investigates the model using field data from 160 teams of 1081 individuals. The results show that IT enables ambidextrous team process capabilities, with both process standardization and agility having positive influences on the inter-team coordination and team innovation, which in turn lead to a higher team performance. By highlighting the importance of process ambidexterity through the potential nomological net from IT to team performance, this study provides further insightful implications for academia and practitioners in the field of team process ambidexterity and information technology.

Appendix A

Table A1

Table A1

Survey items.

IT Support (Lee & Choi, 2003)	
ITS1	Our team is provided with IT support for collaborative work regardless of time and place.
ITS2	Our team is provided with IT support for communicating among team members.
ITS3	Our team is provided with IT support for searching and accessing necessary information.
ITS4	Our team is provided with IT support for systematic storing.
Process Agility (Lee et al., 2013)	
PA1	Our team is able to sense user requirements changes effectively.
PA2	Our team is able to strategize its response to user requirements changes effectively.
PA3	Our team makes effective decisions to cope with business changes.
PA4	Our team effectively performs the changing business requirements.
Process Standardization (Lee et al., 2013)	
PS1	Our team consistently use common task processes across the team.
PS2	Our team consistently use common planning methods/techniques across the team.
PS3	Our team consistently use common communication methods/technologies across the team.
PS4	Our team consistently use common performance review methods/processes across the team.
Team Innovation (Lovelace, Shapiro, and Weingart (2001))	
TI1	Our team constantly develops new products and services for new market.
TI2	Our team collects information regarding new products and markets.
TI3	Our team develops capabilities to improve the current business model.
TI4	Our team develops technologies creating new business opportunities.
Inter-team Coordination (Hoegl et al., 2004)	
IC1*	Duplicated and overlapping coordination activities were avoided.
IC2	We had no problems in coordinating with other teams.
IC3	Conflicts with other teams were settled quickly.
IC4	Discussions with other teams were conducted constructively.
IC5*	Our team is able to adjust processes with respect to coordination with other teams.
Team Performance (Hoegl et al., 2004)	
TP1*	Going by the current status, our team can be regarded as successful.
TP2	All of our team-level goals have been achieved.
TP3	Our team's output so far is of high quality.
TP4	Our team is satisfied with its performance to this point.
TP5	Our team leadership can be fully satisfied with the task progress of this team

* These variables were dropped for the validity of measures.

Appendix B

Table B1

Table B1

Common method bias analysis.

Construct	Indicator	Substantive factor loading (R1)	R1-square	Method factor loading (R2)	R2-square
IT Support	ITS1	0.97 ***	0.94	−0.04	0.00
	ITS2	0.97 ***	0.94	0.00	0.00
	ITS3	0.93 ***	0.86	0.05	0.00
	ITS4	0.97 ***	0.95	0.00	0.00
Process Standardization	PS1	0.77 ***	0.60	0.19	0.04
	PS2	1.03 ***	1.05	−0.08	0.01
	PS3	0.99 ***	0.97	−0.03	0.00
	PS4	0.98 ***	0.97	−0.03	0.00
Process Agility	PA1	1.00 ***	1.01	−0.06	0.00
	PA2	0.96 ***	0.92	0.00	0.00
	PA3	1.00 ***	0.99	−0.03	0.00
	PA4	0.95 ***	0.90	0.02	0.00
Inter-team Coordination	IC2	0.96 ***	0.93	0.01	0.00
	IC3	0.97 ***	0.94	−0.09	0.01
	IC4	0.95 ***	0.91	0.00	0.00
	IC4	0.95 ***	0.91	0.00	0.00
Team Innovation	TI1	0.89 ***	0.79	0.09	0.01
	TI2	0.90 ***	0.81	0.04	0.00
	TI3	1.05 ***	1.10	−0.11 **	0.01
	TI4	0.95 ***	0.90	0.02	0.00
Team Performance	TP2	0.93 ***	0.86	0.05	0.00
	TP3	0.83 ***	0.68	0.15	0.02
	TP4	0.87 ***	0.76	0.10	0.01
	TP5	1.04 ***	1.08	−0.13	0.02
Average			0.91		0.01

** p < 0.01.

*** p < 0.001.

References

- Ali, M., Kan, K. A. S., & Sarstedt, M. (2016). Direct and configurational paths of absorptive capacity and organizational innovation to successful organizational performance. *Journal of Business Research*, 69(11), 5317–5323.
- Anderson, N., Potočník, K., & Zhou, J. (2014). Innovation and creativity in organizations a state-of-the-science review, prospective commentary, and guiding framework. *Journal of Management*, 40(5), 1297–1333.
- Athey, S., & Schmutzler, A. (1995). Product and process flexibility in an innovative environment. *The Rand Journal of Economics*, 557–574.
- Bain, P. G., Mann, L., & Pirola-Merlo, A. (2001). The innovation imperative: The relationships between team climate, innovation, and performance in research and development teams. *Small Group Research*, 32(1), 55–73.
- Bala, H., Massey, A. P., & Montoya, M. M. (2017). The effects of process orientations on collaboration technology use and outcomes in product development. *Journal of Management Information Systems*, 34(2), 520–559.
- Beimborn, D., Gleisner, F., Joachim, N., & Hackethal, A. (2009). The role of process standardization in achieving it business value. *2009 42nd Hawaii International Conference on System Sciences*, 1–10.
- Benner, M. J., & Tushman, M. (2002). Process management and technological innovation: A longitudinal study of the photography and paint industries. *Administrative Science Quarterly*, 47(4), 676–707.
- Cha, J., Kim, Y., Lee, J.-Y., & Bachrach, D. G. (2015). Transformational leadership and inter-team collaboration: Exploring the mediating role of teamwork quality and moderating role of team size. *Group & Organization Management*, 40(6), 715–743.
- Chen, Y., Wang, Y., Nevo, S., Jin, J., Wang, L., & Chow, W. S. (2014). It capability and organizational performance: The roles of business process agility and environmental factors. *European Journal of Information Systems*, 23(3), 326–342.
- Chi, M., Zhao, J., George, J. F., Li, Y., & Zhai, S. (2017). The influence of inter-firm it governance strategies on relational performance: The moderation effect of information technology ambidexterity. *International Journal of Information Management*, 37(2), 43–53.
- Chin, W. W. (1998). The partial least squares approach to structural equation modeling. *Modern Methods for Business Research*, 295(2), 295–336.
- Choi, J. N. (2007). Change-oriented organizational citizenship behavior: Effects of work environment characteristics and intervening psychological processes. *Journal of Organizational Behavior*, 28(4), 467–484.
- Choi, J. N., Anderson, T. A., & Veillette, A. (2009). Contextual inhibitors of employee creativity in organizations the insulating role of creative ability. *Group & Organization Management*, 34(3), 330–357.
- Choi, J. N., & Chang, J. Y. (2009). Innovation implementation in the public sector: An integration of institutional and collective dynamics. *The Journal of Applied Psychology*, 94(1), 245.
- Choi, K., Narasimhan, R., & Kim, S. W. (2016). Opening the technological innovation black box: The case of the electronics industry in Korea. *European Journal of Operational Research*, 250(1), 192–203.
- Choi, S. Y., Lee, H., & Yoo, Y. (2010). The impact of information technology and transactive memory systems on knowledge sharing, application, and team performance: A field study. *MIS Quarterly*, 34(4), 855–870.
- Coltman, T., Tallon, P., Sharma, R., & Queiroz, M. (2015). *Strategic it alignment: Twenty-five years on*. London, England: SAGE Publications Sage UK.
- Davison, R. B., Hollenbeck, J. R., Barnes, C. M., Sleesman, D. J., & Ilgen, D. R. (2012). Coordinated action in multiteam systems. *The Journal of Applied Psychology*, 97(4), 808.
- de Guinea, A. O., & Raymond, L. (2020). Enabling innovation in the face of uncertainty through it ambidexterity: A fuzzy set qualitative comparative analysis of industrial service Smes. *International Journal of Information Management*, 50, 244–260.
- de Vries, T. A., Walter, F., Van der Vegt, G. S., & Essens, P. J. (2014). Antecedents of individuals' interteam coordination: Broad functional experiences as a mixed blessing. *The Academy of Management Journal*, 57(5), 1334–1359.
- Edmondson, A. C. (2002). The local and variegated nature of learning in organizations: A group-level perspective. *Organization Science*, 13(2), 128–146.
- Fixson, S. K., & Park, J.-K. (2008). The power of integrality: Linkages between product architecture, innovation, and industry structure. *Research Policy*, 37(8), 1296–1316.
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50.
- Funk, J. L., & Luo, J. (2015). Open standards, vertical disintegration and entrepreneurial opportunities: How vertically-specialized firms entered the us semiconductor industry. *Technovation*, 45, 52–62.
- Gattiker, T. F., Chen, D., & Goodhue, D. L. (2005). Agility through standardization: A Crm/Erp application. 2005 In E. Bendoly, & F. R. Jacobs (Eds.). *Strategic ERP extension and use* (pp. 87–96). Stanford: Stanford University Press.
- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2006). *Multivariate data analysis*, Vol. 6. Upper Saddle River, NJ: Pearson Prentice Hall.
- He, Z.-L., & Wong, P.-K. (2004). Exploration Vs. exploitation: An empirical test of the ambidexterity hypothesis. *Organization Science*, 15(4), 481–494.
- Hoegl, M., Weinkauff, K., & Gemuenden, H. G. (2004). Interteam coordination, project commitment, and teamwork in multiteam R&D projects: A longitudinal study.

- Organization Science, 15(1), 38–55.
- Huang, K.-E., Wu, J.-H., Lu, S.-Y., & Lin, Y.-C. (2016). Innovation and technology creation effects on organizational performance. *Journal of Business Research*, 69(6), 2187–2192.
- Huang, P.-Y., Pan, S. L., & Ouyang, T. H. (2014). Developing information processing capability for operational agility: Implications from a Chinese manufacturer. *European Journal of Information Systems*, 23(4), 462–480.
- James, L. R., Demaree, R. G., & Wolf, G. (1984). Estimating within-group interrater reliability with and without response bias. *The Journal of Applied Psychology*, 69(1), 85–98.
- Joglekar, N. R., Yassine, A. A., Eppinger, S. D., & Whitney, D. E. (2001). Performance of coupled product development activities with a deadline. *Management Science*, 47(12), 1605–1620.
- Kohli, R., & Grover, V. (2008). Business value of it: An essay on expanding research directions to keep up with the times. *Journal of the Association for Information Systems*, 9(2), 23–39.
- Lee, G., DeLone, W. H., & Espinosa, J. A. (2006). Ambidextrous coping strategies in globally distributed software development projects. *Communications of the ACM*, 49(10), 35–40.
- Lee, G., DeLone, W. H., & Espinosa, J. A. (2010). *The main and interaction effects of process rigor, process standardization, and process agility on system performance in distributed is development: An ambidexterity perspective*. ICIS pp. 34.
- Lee, G., Espinosa, J. A., & DeLone, W. H. (2013). Task environment complexity, global team dispersion, process capabilities, and coordination in software development. *IEEE Transactions on Software Engineering*, 39(12), 1753–1771.
- Lee, H., & Choi, B. (2003). Knowledge management enablers, processes, and organizational performance: An integrative view and empirical examination. *Journal of Management Information Systems*, 20(1), 179–228.
- Lee, J., Min, J., Kwak, C., Pee, L., & Lee, H. (2019). Share or send and receive? The impact of team knowledge outflow/inflow with its support on performance. *Journal of Knowledge Management*. <https://doi.org/10.1108/JKM-11-2018-0705> Vol. ahead-of-print No. ahead-of-print.
- Lee, J., Min, J., & Lee, H. (2017). Setting a knowledge boundary across teams: Knowledge protection regulation for inter-team coordination and team performance. *Journal of Knowledge Management*, 21(2), 254–274.
- Liang, H., Saraf, N., Hu, Q., & Xue, Y. (2007). Assimilation of enterprise systems: The effect of institutional pressures and the mediating role of top management. *MIS Quarterly*, 31(1), 59–87.
- Lovelace, K., Shapiro, D. L., & Weingart, L. R. (2001). Maximizing cross-functional new product teams' innovativeness and constraint adherence: A conflict communications perspective. *The Academy of Management Journal*, 44(4), 779–793.
- Lyytinen, K., & Rose, G. M. (2006). Information system development agility as organizational learning. *European Journal of Information Systems*, 15(2), 183–199.
- Manders, B., de Vries, H. J., & Blind, K. (2016). Iso 9001 and product innovation: A literature review and research framework. *Technovation*, 48, 41–55.
- Mao, H., Liu, S., Zhang, J., & Deng, Z. (2016). Information technology resource, knowledge management capability, and competitive advantage: The moderating role of resource commitment. *International Journal of Information Management*, 36(6), 1062–1074.
- March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87.
- Marks, M. A., DeChurch, L. A., Mathieu, J. E., Panzer, F. J., & Alonso, A. (2005). Teamwork in multiteam systems. *The Journal of Applied Psychology*, 90(5), 964–971.
- Marrone, J. A., Tesluk, P. E., & Carson, J. B. (2007). A multilevel investigation of antecedents and consequences of team member boundary-spanning behavior. *The Academy of Management Journal*, 50(6), 1423–1439.
- Mathiasen, L., & Pries-Heje, J. (2006). Business agility and diffusion of information technology. *European Journal of Information Systems*, 15(2), 116–119.
- Melville, N., Kraemer, K., & Gurbaxani, V. (2004). Review: Information technology and organizational performance: An integrative model of its business value. *MIS Quarterly*, 28(2), 283–322.
- Mithas, S., Ramasubbu, N., & Sambamurthy, V. (2011). How information management capability influences firm performance. *MIS Quarterly*, 35(1), 237.
- Münstermann, B., Eckhardt, A., & Weitzel, T. (2010). The performance impact of business process standardization: An empirical evaluation of the recruitment process. *Business Process Management Journal*, 16(1), 29–56.
- Olabisi, J., & Lewis, K. (2018). Within- and between-team coordination via transactive memory systems and boundary spanning. *Group & Organization Management*, 43(5), 691–717.
- Park, Y., El Sawy, O. A., & Fiss, P. C. (2017). The role of business intelligence and communication technologies in organizational agility: A configurational approach. *Journal of the Association for Information Systems*, 18(9), 648–686.
- Peng, J., Quan, J., Zhang, G., & Dubinsky, A. J. (2016). Mediation effect of business process and supply chain management capabilities on the impact of it on firm performance: Evidence from Chinese firms. *International Journal of Information Management*, 36(1), 89–96.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies. *Journal of Applied Psychology*, 88(5), 879–903.
- Podsakoff, P. M., & Organ, D. W. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management*, 12(4), 531–544.
- Queiroz, M., Tallon, P. P., Sharma, R., & Colman, T. (2018). The role of its application orchestration capability in improving agility and performance. *The Journal of Strategic Information Systems*, 27(1), 4–21.
- Raisch, S., & Birkinshaw, J. (2008). Organizational ambidexterity: Antecedents, outcomes, and moderators. *Journal of Management*, 34(3), 375–409.
- Ramakumar, A., & Cooper, B. (2004). Process standardization proves profitable. *Quality*, 43(2), 42–45.
- Raschke, R. L. (2010). Process-based view of agility: The value contribution of it and the effects on process outcomes. *International Journal of Accounting Information Systems*, 11(4), 297–313.
- Ray, G., Barney, J. B., & Muhanna, W. A. (2004). Capabilities, business processes, and competitive advantage: Choosing the dependent variable in empirical tests of the resource-based view. *Strategic Management Journal*, 25(1), 23–37.
- Reif, J. A., Kugler, K. G., & Brodbeck, F. C. (2018). The regulatory power of standardized business processes. *Business Process Management Journal*.
- Rosemann, M. (2014). *Proposals for future bpm research directions*. Asia-Pacific conference on business process management. Springer1–15.
- Rosemann, M., & vom Brocke, J. (2015). *The six core elements of business process management*. Handbook on business process management, 1, Springer105–122.
- Ross, J. W., & Vitale, M. R. (2000). The erp revolution: Surviving Vs. Thriving. *Information Systems Frontiers*, 2(2), 233–241.
- Sambamurthy, V., Bharadwaj, A., & Grover, V. (2003). Shaping agility through digital options: Reconceptualizing the role of information technology in contemporary firms. *MIS Quarterly*, 237–263.
- Sethi, R. (2000). New product quality and product development teams. *Journal of Marketing*, 64(2), 1–14.
- Shin, S. J., Kim, T.-Y., Lee, J.-Y., & Bian, L. (2012). Cognitive team diversity and individual team member creativity: A cross-level interaction. *The Academy of Management Journal*, 55(1), 197–212.
- Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2), 420–428.
- Shukla, S. K. (2020). Evaluating the practices of flexibility maturity for the software product and service organizations. *International Journal of Information Management*, 50, 71–89.
- Sidorova, A., Torres, R., & Al Bayez, A. (2015). *The role of information technology in business process management*. Handbook on business process management, 1, Springer421–444.
- Taggar, S. (2002). Individual creativity and group ability to utilize individual creative resources: A multilevel model. *The Academy of Management Journal*, 45(2), 315–330.
- Talke, K., Salomo, S., & Kock, A. (2011). Top management team diversity and strategic innovation orientation: The relationship and consequences for innovativeness and performance. *The Journal of Product Innovation Management*, 28(6), 819–832.
- Tallon, P. P. (2008). Inside the adaptive enterprise: An information technology capabilities perspective on business process agility. *Information Technology and Management*, 9(1), 21–36.
- Tallon, P. P., & Pinsonneault, A. (2011). Competing perspectives on the link between strategic information technology alignment and organizational agility: Insights from a mediation model. *MIS Quarterly*, 35(2), 463–486.
- Tregear, R. (2015). *Business process standardization*. Handbook on business process management, 2, Springer421–441.
- Trinh, P., Molla, A., & Peszynski, K. (2013). Enterprise systems and organizational agility: A review of the literature and conceptual framework. *Communications of the Association for Information Systems*, 31(1), 167–193.
- Trkman, P. (2010). The critical success factors of business process management. *International Journal of Information Management*, 30(2), 125–134.
- Van Oosterhout, M., Waarts, E., & van Hillegersberg, J. (2006). Change factors requiring agility and implications for it. *European Journal of Information Systems*, 15(2), 132–145.
- vom Brocke, J., Zelt, S., & Schmiedel, T. (2016). On the role of context in business process management. *International Journal of Information Management*, 36(3), 486–495.
- Wright, C., Sturdy, A., & Wylie, N. (2012). Management innovation through standardization: Consultants as standardizers of organizational practice. *Research Policy*, 41(3), 652–662.
- Wüllenweber, K., Beimborn, D., Weitzel, T., & König, W. (2008). The impact of process standardization on business process outsourcing success. *Information Systems Frontiers*, 10(2), 211–224.
- Xie, R., Ling, H., & Zhang, C. (2011). Effect of business process management on firm performance: An ambidexterity perspective. *2011 International Conference on Business Management and Electronic Information*, 3, 341–345.
- Yunis, M., Tarhini, A., & Kassab, A. (2018). The role of ICT and innovation in enhancing organizational performance: The catalysing effect of corporate entrepreneurship. *Journal of Business Research*, 88, 344–356.